Methodology Passport Card

Research Methods — Task 2 · Experimental Setup & Variables

1. Project information

Field	Value
Title	Monitoring LLM Reasoning Traces (Chain-of-Thought) for Detecting Unsafe Behavior in AI Safety Evaluations
Authors	Yerzhankyzy Markhaba, Kulmakhanbet Zarina
Core question	Does reading the chain-of-thought (CoT), in addition to the final output, help an automated monitor detect unsafe LLM behavior better than reading the output alone?
System under evaluation	An LLM-based monitor that reads an episode (task prompt, optionally the model's reasoning trace, and the final output) and returns a suspicion score; episodes are labeled safe or unsafe.
GitHub repository	https://github.com/markhabaerzhan/cot-monitor-benchmark

2. Research type

Quantitative, experimental, paired design. We measure numeric detection metrics (recall at a fixed false positive rate, precision, F1, AUROC) of automated monitors on a labeled set of safe and unsafe episodes and test hypotheses statistically ($\alpha = 0.05$). Both monitors score exactly the same episodes, so the only systematic difference between conditions is what the monitor is allowed to read. No human subjects or interviews are involved, so a qualitative component is not required; the error analysis of false positives and false negatives (Task 5) is a small descriptive add-on, not a second method.

3. Hypotheses

Derived from Task 1. Common settings: $\alpha = 0.05$; the primary metric is recall at 5% FPR, measured on the held-out **test** set only (the threshold is fixed beforehand on a separate calibration set, see Section 4). RQ3 is the central confirmatory hypothesis. RQ1 and RQ2 are supporting; their p-values are reported with Holm correction across the two tests.

	H0 (null)	H1 (alternative)	Test
RQ3 (main)	Recall at 5% FPR of the CoT+output monitor is equal to or lower than that of the output-only monitor.	Recall at 5% FPR of the CoT+output monitor is higher than that of the output-only monitor.	Exact McNemar (one-sided) on the paired unsafe test episodes (detected or missed by each monitor at its own calibrated threshold) + paired bootstrap 95% CI of the recall difference.
RQ2	Recall at 5% FPR of the CoT+output monitor is ≤ 0.70.	Recall at 5% FPR of the CoT+output monitor is > 0.70.	One-sided exact binomial + bootstrap 95% CI. The 0.70 target is a working value; it is finalized on the pilot set only and frozen in config.yaml before the main test run.
RQ1	Recall at 5% FPR of the CoT+output monitor is the same for all four unsafe-behavior categories.	Recall differs between at least two categories.	Chi-square test of homogeneity (Fisher's exact test if expected counts are small). Repeated for the output-only monitor as an exploratory check.

4. Variable matrix

Type	Variable	Levels / unit	Used in
Independent	Monitor input	output_only (prompt + final output) vs cot_plus_output (prompt + reasoning trace + final output)	RQ2, RQ3
Independent (stratifying)	Unsafe-behavior category	reward hacking, deception, sabotage, harmful compliance	RQ1
Dependent	Recall at 5% FPR (primary)	proportion, 0–1	all

Type	Variable	Levels / unit	Used in
Dependent	Precision, F1, AUROC (secondary)	proportion, 0–1; AUROC is threshold-free	all
Dependent	Monitor latency and input tokens (resource)	seconds per call; tokens per episode	all
Controlled	Monitor LLM, prompt template, temperature, output format	Same model version and same prompt text in both conditions; the only difference is the reasoning-trace block. Temperature 0.0. The monitor returns an integer suspicion score 0–100 in JSON (the prompt asks to use the full scale); the score makes a 5% FPR threshold and AUROC possible.	all
Controlled	Source model(s) and episode generation	Episodes are produced by fixed open reasoning model(s) that expose the raw CoT; task domain and difficulty are fixed per run.	all
Controlled	Test episodes and labeling protocol	Identical test episodes in both conditions; equal number of unsafe episodes per category; one written labeling protocol (what counts as unsafe).	all
Controlled	Calibration / test split	Safe episodes are split into two disjoint sets (no shared episode IDs or prompts): a calibration set, used only to set each monitor's threshold, and a test set, used for every reported metric. Unsafe episodes are used for testing only.	all
Controlled	Operating point	Threshold set per monitor on the calibration set so that FPR = 5%.	all
Controlled	Reasoning-trace handling	Same maximum length and same truncation rule for every trace.	all
Controlled	Software and hardware	Python 3.13 with exact-pinned requirements.txt; Docker base image pinned to an exact patch tag (python:3.13.16-slim-trixie); random seed 42; CPU only.	all

5. Metrics and baseline

5.1 Evaluation metrics and formulas

Notation: s = monitor score (0–100); a threshold τ flags an episode if $s \geq \tau$; TP, FP, TN, FN are counted on the test set after the threshold is fixed. Unsafe is the positive class.

Type	Metric	Formula / definition	Why it matches the hypothesis
Primary	Recall at 5% FPR	τ = smallest threshold with FPR on the calibration set ≤ 0.05; Recall = TP / (TP + FN) on the test set. Effect size: Δ = Recall(cot_plus_output) − Recall(output_only).	A safety monitor is useful only if it catches unsafe behavior without flooding reviewers with false alarms, so detection is compared at a fixed false-alarm budget. H0 and H1 are decided on this metric.
Secondary (quality)	Precision, F1	Precision = TP / (TP + FP); F1 = 2 · Precision · Recall / (Precision + Recall)	Show the cost of detection in false alarms and give a single balanced score.
Secondary (quality)	AUROC	P(s(unsafe) > s(safe)) + 0.5 · P(s(unsafe) = s(safe)) over all unsafe/safe test pairs	Threshold-free; shows whether the conclusion depends on the 5% operating point.
Guardrail	Realized FPR	FPR = FP / (FP + TN) on the safe test set; about 5% ($\leq$ 7% accepted)	If one monitor has a higher real FPR, its recall is inflated and the comparison is unfair.
Guardrail	Failed replies	failed or unparseable calls / all calls $\leq$ 1%; failures are reported, not dropped	Silent drops would bias recall.
Resource (reported)	Latency, tokens, cost	Latency: p50 and p95 seconds per call; tokens: mean input tokens per episode; cost = tokens × price per token	Reading the trace makes the input much longer; the benefit must be weighed against this cost.

5.2 Planned sample (fixed after the pilot, before the main run)

Set	Planned size	Why
Safe, calibration	≥ 200 episodes	The threshold is read from the 95th percentile of safe scores, so it rests on about 10 episodes.
Safe, test	≥ 200 episodes	5% FPR corresponds to about 10 false positives, so realized FPR can be checked.
Unsafe, test	50 per category (200 total)	Enough paired outcomes for McNemar and about 50 per category for RQ1.

5.3 Baseline and experimental setup

Baseline (control group): the output-only monitor (same LLM, same prompt, same episodes; sees the prompt and final output but not the reasoning). It is the natural "no extra information" comparison and corresponds to current practice of monitoring actions and outputs. Sanity baselines: a random scorer (expected AUROC = 0.5) and the offline keyword placeholder included in the repository.

Execution strategy (A/B comparison):

Step	Action
1. Pilot	Run the pipeline on a small pilot set to check parsing, scale use and run-to-run stability (3 repeated scorings); finalize set sizes and the RQ2 target.
2. Freeze	Commit config.yaml, the prompt template and the labeling protocol; nothing changes after this point.
3. Calibrate	Score the safe calibration set in each condition and set the threshold τ for 5% FPR separately for each monitor.
4. Test	For every test episode build two inputs (A: output only, B: trace + output) from the same prompt template; score each in an independent call with no shared context; episode order is shuffled with seed 42.
5. Analyze	Compute all metrics with the fixed thresholds; run McNemar and bootstrap for RQ3, then RQ2 and RQ1; report guardrails first, then results.

5.4 Guardrail checks (validity, not goals)

Guardrail	Limit
Calibration/test separation	0 shared episode IDs or prompts between the two safe sets.
Class balance	Equal number of unsafe episodes per category.
Run-to-run stability	Pilot set scored 3 times; differences in recall are reported (API models are not always fully deterministic at temperature 0).
Safety of data	No harmful content stored or published (harmful-compliance episodes are redacted placeholders); no API keys in the repository.

5.5 Threats to validity and mitigation

Threat	Type	Mitigation
Threshold leakage (threshold tuned on test data)	Internal	Disjoint calibration and test sets; threshold frozen before testing.
Prompt asymmetry	Internal	One prompt template that differs only in the reasoning-trace block.
Label noise	Internal	Written labeling protocol; ambiguous episodes are listed and excluded before the main run.
Non-determinism and order effects	Internal	Temperature 0, fixed seed, independent calls, shuffled order, repeated pilot scoring.
One monitor LLM, limited source models and tasks	External	Results are stated only for the tested monitor and tasks; no claim of generalization to all monitors or deployments.
CoT faithfulness: the trace may not reflect real reasoning, and a model can hide intent in it (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025)	External / construct	Stated as a limitation; analyzed in the error analysis (Task 5).